

Class 11: Structural Bioinformatics pt2

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Background

We saw last that the PDB has 209,886 entries (Oct/Nov 2025). UniProtKB (i.e. protein sequence database) has 199,579,901 entries.

```
209886/199579901 * 100
```

```
[1] 0.1051639
```

So the PDB has only 0.1% coverage of the main sequence database.

Enter AlphaFold data base (AFDB) < <https://alphafold.ebi.ac.uk> > that attempts to provide computed models for all sequences in UniProt.

“AlphaFold DB provides open access to over 200 million protein structure predictions to accelerate scientific research.”

AlphaFold

AlphaFold has 3 main outputs

- the predicted coordinates (PDB files)
- A local quality score called **pLDDT** (one for each amino acid)
- A second quality score **PAE** Predicted Aligned Error (for each pair of amino acid)

We can run AlphaFold ourselves if we are not happy with AFDB (i.e. no coverage or poor model).

Interpreting/analyzing AF results in R

```
results_dir <- "HIVPR_dimer_23119/"
```

```
pdb_files <- list.files(path=results_dir,  
                        pattern="*.pdb",  
                        full.names = TRUE)
```

```
# Print our PDB file names  
basename(pdb_files)
```

```
[1] "HIVPR_dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb"  
[2] "HIVPR_dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb"  
[3] "HIVPR_dimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb"  
[4] "HIVPR_dimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb"  
[5] "HIVPR_dimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb"
```

```
library(bio3d)  
library(bio3d.view)  
pdbs <- pdbaln(pdb_files, fit=TRUE, exefile="msa")
```

Reading PDB files:

```
HIVPR_dimer_23119//HIVPR_dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_0  
HIVPR_dimer_23119//HIVPR_dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_0  
HIVPR_dimer_23119//HIVPR_dimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_0  
HIVPR_dimer_23119//HIVPR_dimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_0  
HIVPR_dimer_23119//HIVPR_dimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_0  
.....
```

Extracting sequences

```
pdb/seq: 1   name: HIVPR_dimer_23119//HIVPR_dimer_23119_unrelaxed_rank_001_alphafold2_multime  
pdb/seq: 2   name: HIVPR_dimer_23119//HIVPR_dimer_23119_unrelaxed_rank_002_alphafold2_multime  
pdb/seq: 3   name: HIVPR_dimer_23119//HIVPR_dimer_23119_unrelaxed_rank_003_alphafold2_multime  
pdb/seq: 4   name: HIVPR_dimer_23119//HIVPR_dimer_23119_unrelaxed_rank_004_alphafold2_multime  
pdb/seq: 5   name: HIVPR_dimer_23119//HIVPR_dimer_23119_unrelaxed_rank_005_alphafold2_multime
```

```
#library(rgl)  
library(bio3d)  
library(bio3d.view)
```

```
view.pdbs(pdbs)
```

```
aln_file <- list.files(path=results_dir,  
                      pattern=".a3m$",  
                      full.names = TRUE)  
aln_file
```

```
[1] "HIVPR_dimer_23119//HIVPR_dimer_23119.a3m"
```

```
aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

```
[1] " ** Duplicated sequence id's: 101 **"  
[2] " ** Duplicated sequence id's: 101 **"
```

```
sim <- conserv(aln)
```